

# Camilo Laiton

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## Summary

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Software Engineer specializing in deep learning and cloud computing. Experienced in designing and deploying scalable image processing and data analysis pipelines, with strong experience working in collaborative teams and mentoring junior engineers. Proven ability to enhance large-scale data analysis, achieving significant improvements in processing times and accuracy through innovative solutions and collaborative efforts.

## Online Profiles

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- [Portfolio](#)
- [Github](#)
- [Linkedin](#)

## Experience

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### Allen Institute | Seattle, WA

#### Software Engineer III - Computer Vision Engineer | 10/2023 - Present

- Retrospective flat-field correction applied to large-scale data (hundreds of GBs to many TBs).
- Maintained a cloud image processing pipeline which processes around 144 TBs a month.
- Conducted code reviews to ensure quality and adherence to best practices.
- Mentored junior engineers on software development methodologies and techniques.
- Collaborated with scientists to define software requirements and specifications.
- Development of a template to register brains acquired from lightsheet imaging to the Allen Common Coordinate Framework (CCF V3).
- Large-scale mRNA identification using cloud computing resources.
- Large-scale cell segmentation with deep learning, using cloud computing resources.
- Development of a Python package to perform fast image processing operations in the cloud.
- Deep learning model for brain ventricle segmentation to improve the registration process to the Allen CCF atlas.
- Deep learning model for brain tissue segmentation to improve processing times in large-scale datasets.
- Development of a framework for fast, large-scale cell detection (ZYG positions) in brains acquired with lightsheet microscopes.
- Writing Nextflow scripts and deployment of Apptainer/Singularity compatible OCI images for image processing pipelines in the cloud and in SLURM clusters.
- **Currently working on:** a foundational deep learning model for brain imaging data from lightsheet microscopy and a deep learning model for anti-Bassoon and anti-PSD95 protein prediction from the NHS-ester (pan-protein) channel.

### Allen Institute | Seattle, WA

#### Software Engineer II - Computer Vision Engineer | 07/2022 - 10/2023

- Development of image processing pipelines for light-sheet microscope images using NextFlow and AWS.
- Some of the image processing steps are: destripping, 3D reconstruction, format image conversion to OME-Zarr, atlas registration, and explorative analysis on cell segmentation and mRNA detection.
- Development of a bioimage analysis software package to estimate stitching errors in microscopy datasets.
- Uploading data to AWS using a HPC cluster.
- Working with large-scale data from hundreds of GBs to many TBs from very high-resolution microscopes.

### Institución Universitaria ITSA | Barranquilla, ATL

#### Software Engineer | 04/2021 - 07/2022

- Reviewed the state of the art tools for the segmentation of brain anatomical structures.
- Installation of tools on Linux environments.
- Developed the intermediary module in Python for the automatic use of brain image segmentation tools.

### COTECMAR | Cartagena, BOL

#### Research Associate | 11/2021 - 04/2022

- Provided support and analysis of dual-use technologies incorporating electromobility technologies available in the Colombian and international market.

- Development of simulation software using evolutionary algorithms to reduce time, pollution and costs by using a transport route with multimodal nodes (river, rail and road) for the Colombian ferrofluvial 4.0 project.

## **SIE SOLUCIONES IT | Santa Marta, MAG**

**Web Developer | 07/2020 - 05/2021**

- Development of a web platform based on non-immersed virtual reality for the optimization of time and resources of the company Ladys Confecciones S.A.S.

## **Universidad Nacional de Colombia | Medellín, ANT**

**Data Engineer | 09/2020 - 03/2021**

- Implementation of a Geoscientific Data Lake for the Servicio Geológico Colombiano (SGC).
- Implemented ETL processes for automated loading of large volumes of data into the data lake.
- Implemented performance tuning techniques on mappings, sessions and workflows to improve throughput rate of ETL process.

## **Skills**

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Deep learning (PyTorch, TensorFlow), Cloud computing (AWS, Azure), Data Analysis, Large-scale data processing, Design, deployment and optimization of automated pipelines, Software Development

## **Education**

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### **Universidad Nacional de Colombia | Medellín**

**Masters in Sciences, Computer Science | 12/2022**

- Honorary diploma meritorious degree.
- BEDA Scholarship Waiver Academic Rights for Academic Performance.
- 5.0/5.0 GPA.
- Published thesis: [Deep 3D Neural Network for Brain Structures Segmentation Using Self-Attention Modules in MRI Images.](#)

### **Universidad del Magdalena | Santa Marta, CO**

**System's engineering | 12/2020**

- Graduated Summa Cum Laude.
- 4.5/5.0 GPA.
- Honorific graduate degree for excellent performance in undergraduate thesis.
- Published thesis: A CNN-based model for dynamic traffic signal timing estimation at simple urban intersections.

## **Languages**

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English, Spanish, French

## **Certificates**

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DeepLearning.AI TensorFlow Developer Specialization., Creating Multi-Task Models with Keras, Delf B2

## **Publications**

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- [SmartSPIM Pipeline: A Scalable Cloud-Based Image Processing Pipeline for Light-sheet Microscopy Data](#)
- [Cloud Pipelines for Large Scale Lightsheet Image Processing](#)
- [Deep 3D neural network for brain structures segmentation using self-attention modules in MRI images](#)
- [Industry 4.0 Technologies Applied to the Rail Transportation Industry: A Systematic Review](#)
- [A CNN-based model for dynamic traffic signal timing estimation at simple urban intersections](#)